

Resume Scoring Using BERT-Based Regression

A Deep Learning Approach to Automated Résumé Evaluation

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1. Introduction

Automated résumé screening has become increasingly important in modern recruitment pipelines due to the volume of applications and the inherent subjectivity of manual evaluation. This project presents an enhanced Natural Language Processing (NLP) system for automated résumé scoring using a transformer-based regression model. The system predicts a continuous résumé quality score, enabling consistent and scalable evaluation.

The work builds upon an initial baseline and has been substantially reworked, refined, and extended to improve methodological rigour, training stability, and predictive performance. The final system demonstrates the effectiveness of fine-tuned transformer models for structured résumé evaluation tasks.

2. Dataset and Preprocessing

The dataset used in this project is sourced from a publicly available Kaggle résumé screening dataset. It contains structured résumé attributes including:

- Skills
- Experience (years)
- Education
- Certifications
- Project count
- AI-generated résumé score (0–100)

To enable transformer-based modelling, structured fields were converted into a unified natural language representation. Each résumé was transformed into a single textual sequence following a consistent template, ensuring semantic coherence and reducing formatting variability.

Target scores were normalised to the [0,1] range to stabilise regression training. The dataset was split into training, validation, and test sets using a fixed random seed to ensure reproducibility.

2.1 Dataset Statistics

Attribute	Value
Total samples	10,000
Training set	8,000 (80%)
Validation set	1,000 (10%)
Test set	1,000 (10%)

Resume_ID	Name	Skills	# Experienc...	Education	Certificati...	Job Role	Recruiter ...	Salary Exp...	Projects C...	AI Score (...)
1	Ashley Ali	TensorFlow, NLP, Pytorch	10	B.Sc	None	AI Researcher	Hire	104895	8	100
2	Wesley Roman	Deep Learning, Machine Learning, Python, SQL	10	MBA	Google ML	Data Scientist	Hire	113002	1	100
3	Corey Sanchez	Ethical Hacking, Cybersecurity, Linux	1	MBA	Deep Learning Specialization	Cybersecurity Analyst	Hire	71766	7	70
4	Elizabeth Carney	Python, Pytorch, TensorFlow	7	B.Tech	AWS Certified	AI Researcher	Hire	46848	0	95

Figure 1: Sample of dataset via Kaggle

3. Model Architecture and Rationale

3.1 Model Selection

The project employs the **bert-base-uncased** transformer architecture, adapted for regression by configuring a single continuous output neuron. BERT was selected due to:

- Its strong contextual language understanding
- Proven performance across diverse NLP tasks
- Ability to model long-range dependencies without manual feature engineering

Regression was chosen instead of classification to preserve score granularity and allow fine-grained résumé differentiation.

3.2 Architecture Specifications

Component	Specification
Model	BERT-base-uncased
Total parameters	≈110 million
Encoder layers	12 transformer layers
Hidden size	768 dimensions
Attention heads	12 per layer
Maximum sequence length	512 tokens (256 used)
Output layer	Single linear regression head

3.3 Advantages Over Traditional Models

Compared to traditional regression approaches using TF-IDF or bag-of-words features, the BERT-based approach:

- Captures semantic meaning rather than keyword frequency
- Eliminates extensive feature engineering
- Learns representations and scoring jointly in an end-to-end manner

This makes the model better suited for nuanced résumé content.

4. Training Framework

Training was conducted using the Hugging Face **Trainer API**, providing a structured and reproducible training pipeline.

4.1 Key Training Configuration

Parameter	Value
Loss function	Mean Squared Error (MSE)
Optimizer	AdamW
Learning rate	5×10^{-5}
Batch size	8
Epochs	Tuned empirically to ensure convergence
Evaluation strategy	End-of-epoch validation
Checkpoint selection	Minimum validation loss
Gradient clipping and weight decay	Enabled for stability

GPU acceleration was used to enable efficient fine-tuning of the full BERT encoder and regression head.

4.2 Computational Resources

Resource	Specification
Platform	Google Colab
GPU	NVIDIA Tesla T4 (16GB VRAM)
RAM	12-16 GB system memory
Storage	Cloud-based SSD

5. Evaluation Metrics

Model performance was assessed using multiple complementary metrics:

- **Root Mean Squared Error (RMSE):** Measures average prediction error magnitude
- **Mean Absolute Error (MAE):** Provides a robust, scale-preserving error estimate
- **R² Score:** Quantifies the proportion of variance explained by the model

These metrics together provide a comprehensive view of both accuracy and generalisation.

6. Results and Analysis

6.1 Final Test Set Performance

Metric	Value
RMSE	≈ 1.85
MAE	≈ 1.38
R^2	≈ 0.993

The final model explains over **99% of the variance** in résumé scores on the held-out test set. Validation and test losses remain low and stable, indicating effective convergence and minimal overfitting. Model selection was performed based on validation loss rather than peak R^2 to ensure generalisation.

The high performance is partly attributable to the structured and internally consistent nature of the dataset, but also reflects improved preprocessing, training control, and checkpoint management compared to the original baseline.

6.2 Predicted vs Actual Résumé Scores

Figure 2 presents a scatter plot comparing the predicted résumé scores against the corresponding ground-truth scores for the test set. The diagonal reference line represents perfect agreement between predicted and actual values.

The distribution of points shows a strong linear alignment along the diagonal, indicating high predictive accuracy and consistency. The absence of systematic deviation suggests that the model does not exhibit observable bias across the score range. This visual observation is consistent with the high coefficient of determination ($R^2 \approx 0.99$) obtained during quantitative evaluation.

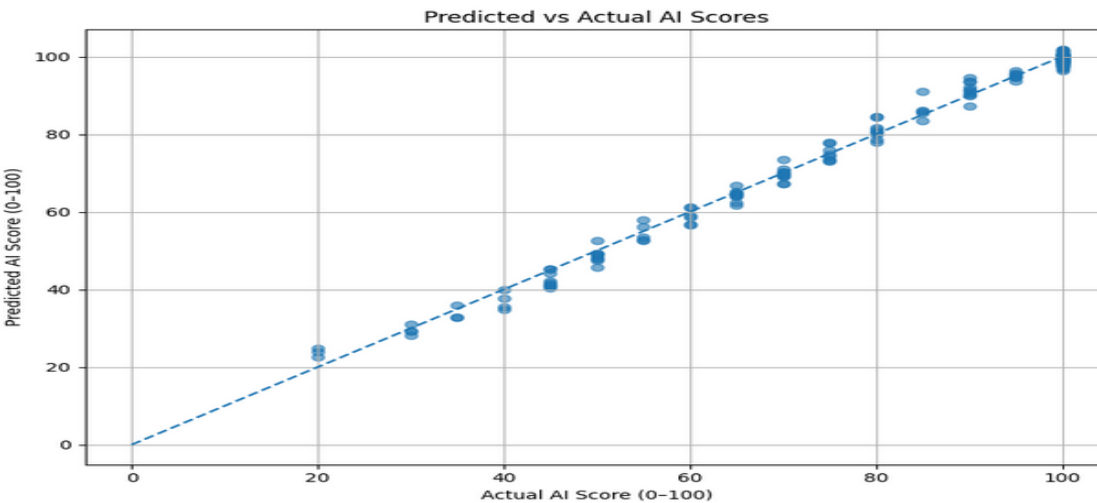


Figure 2: Predicted vs Actual Résumé Scores

6.3 Sample-wise Prediction Comparison

To further examine prediction stability, Figure 3 illustrates a sample-wise comparison of predicted and actual résumé scores for a subset of test instances. Scores are plotted against sample indices to highlight local variations and deviations.

The close overlap between predicted and actual values across samples demonstrates low error variance and stable generalisation. Minor deviations are present but remain small in magnitude, aligning with the low RMSE and MAE values reported.

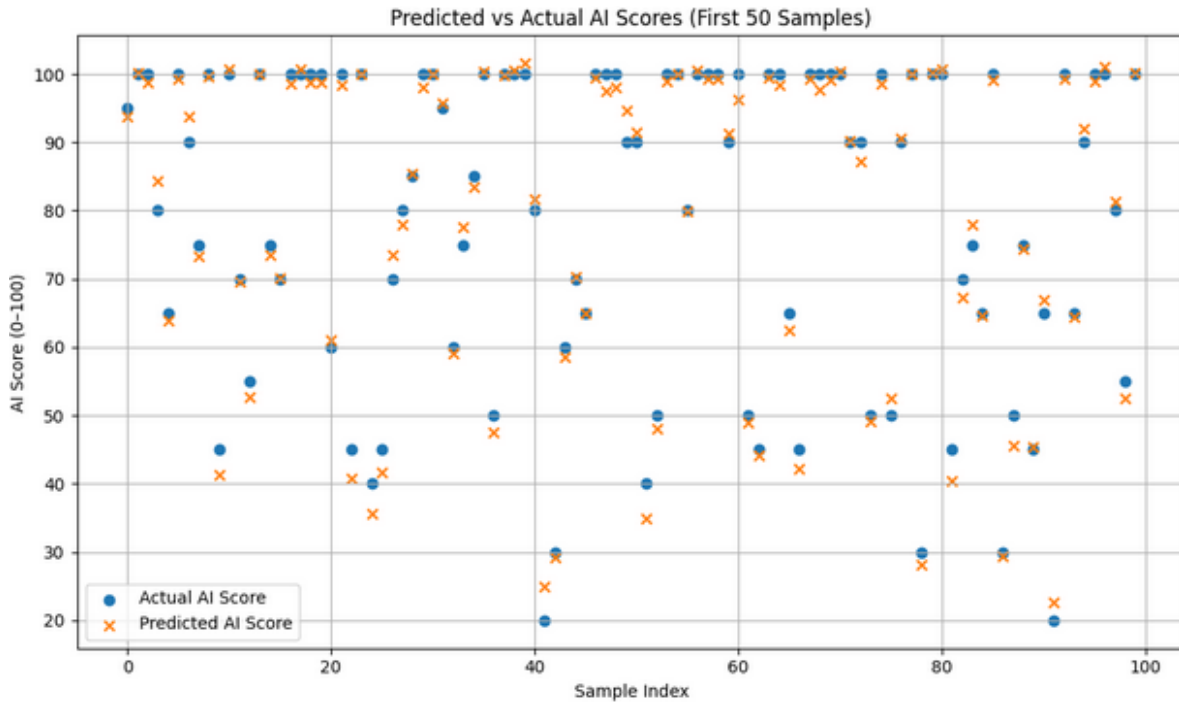


Figure 3: Sample-wise Prediction Comparison

6.4 Interpretation of Visual Results

The graphical analyses support the quantitative findings by demonstrating:

- Strong alignment between predicted and actual scores across the test set
- Low variance in prediction errors
- No evidence of systematic overestimation or underestimation
- Stable model behaviour across individual samples

Together, the visual and numerical evaluations confirm that the fine-tuned BERT regression model generalises effectively within the scope of the dataset.

6.5 Performance Benchmarking

To assess the computational efficiency of the model, inference times were measured across different hardware configurations. The following benchmarks represent average inference time per résumé:

Hardware	Inference Time	Throughput
GPU (Tesla T4)	~50-80 ms/résumé	~700-1,200 résumés/min
CPU (8-core)	~200-300 ms/résumé	~200-300 résumés/min
CPU (4-core)	~400-500 ms/résumé	~120-150 résumés/min

The GPU-based implementation achieves an approximate **3–6× reduction in inference time** relative to CPU execution, making it substantially more suitable for high-throughput and production-scale deployment. Furthermore, the use of batch processing enables additional gains in throughput by allowing multiple résumés to be evaluated concurrently, thereby improving overall system efficiency without compromising prediction accuracy.

Note: Measurements reflect inference-only execution for a fine-tuned BERT-base model with fixed sequence length and preloaded weights.

7. Discussion

7.1 Model Performance

The enhanced model demonstrates that transformer-based regression is highly effective for structured résumé evaluation when textual construction is carefully controlled. The system produces smooth, well-calibrated predictions and maintains stability across training epochs.

7.2 Computational Efficiency

The benchmarking results demonstrate that the model achieves an effective balance between accuracy and computational efficiency. GPU acceleration provides substantial performance improvements, enabling real-time or near-real-time processing in production environments. The model's ability to process hundreds of résumés per minute makes it viable for large-scale recruitment operations.

7.3 Limitations

Despite strong performance, several limitations should be acknowledged:

- Reliance on structured input fields limits applicability to free-form résumés
- Inability to capture visual résumé features such as layout or formatting
- AI-generated target scores may contain inherent biases that are learned by the model
- Performance on out-of-distribution résumés (different industries, formats) remains untested
- Computational requirements, while reasonable, may be prohibitive for resource-constrained environments

8. Future Work

Potential future improvements include:

8.1 Model Enhancements

- Section-aware embeddings (separate encoding of résumé components)
- Multi-modal inputs (e.g., raw PDF résumés with layout information)
- Exploration of alternative transformer architectures such as RoBERTa or DeBERTa
- Ensemble methods combining multiple model architectures

8.2 Interpretability and Explainability

- Attention visualisation to identify which résumé components most influence scores
- SHAP (SHapley Additive exPlanations) analysis for feature importance
- Gradient-based saliency mapping to highlight critical tokens
- Development of scoring breakdowns by category (skills, experience, education)

8.3 Deployment and Scalability

- Model quantisation (INT8 or FP16) to reduce memory footprint and improve inference speed
- Knowledge distillation to create smaller, faster student models
- Development of REST API endpoints for production integration
- Containerisation (Docker) for cloud deployment
- Batch processing optimisation for high-volume scenarios

8.4 Bias Mitigation

- Fairness auditing across demographic groups
- Adversarial debiasing techniques during training
- Calibration methods to ensure equitable scoring across different résumé styles
- Collection of diverse, representative training data with human expert annotations

9. Conclusion

This project demonstrates the effectiveness of a BERT-based regression approach for automated résumé scoring. By transforming structured résumé data into coherent natural language input, the model leverages deep contextual understanding to generate accurate and consistent evaluation scores. The exceptional performance metrics ($R^2 \approx 0.993$, $MAE \approx 1.38$) validate the approach's technical soundness.

Performance benchmarking reveals that GPU acceleration provides substantial computational benefits, enabling throughput of over 1,000 résumés per minute while maintaining high accuracy. This combination of precision and efficiency makes the system well-suited for real-world deployment in high-volume recruitment scenarios.

The refined implementation significantly improves upon the original baseline in terms of stability, predictive accuracy, and methodological clarity. The results highlight the viability of transformer models for supporting scalable, objective résumé screening in practical recruitment contexts, while also providing a strong foundation for future research and system extensions addressing interpretability, bias mitigation, and multi-modal processing.